A Retinal Vessel Segmentation Method Based Improved U-Net Model

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Abstract

There are two problems in retinal blood vessel segmentation, which are the insufficient segmentation of small vessels due to the complex curvature morphology of blood vessels and the segmentation difficulty of blood vessels due to uneven brightness background of lesion fundus images. To solve the problems, a series deformable convolution structure is proposed in this paper, which could improve the adaptive features extraction ability to the blood vessel with various shapes and sizes, enhance feature transmissions and alleviate exploding gradients. On this basis, a retinal vessel segmentation method with series deformable convolution and attention mechanism based on U-net structure(SDAU-Net) is proposed. In SDAU-Net, the convolution module in U-net is replaced by series deformable convolution module, the lightweight attention module and dual attention module are applied in the decoder part, which effectively improves the U-net feature extraction ability for the small vessels with complex morphology and the retinopathy images. To verify the SDAU-Net effect, the comparative experiments are conducted on datasets of DRIVE, STARE, CHASE_DB1 and IOSTAR. The results show that SDAU-Net is superior to comparative methods in accuracy. The SDAU-Net Se and Acc on the DRIVE and STARE are 0.8293, 0.9675, 0.8973 and 0.9833 respectively, which indicate that SDAU-Net has more advantages in small vessel segmentation and lesion images. To verify the generalization and extendibility, cross-dataset and cross-modality experiments are conducted on DRIVE, STARE and IOSTAR, the results demonstrates outstanding generalization and extendibility of SDAU-Net.

*Keywords:*Attention; medical image segmentation; retinal vessel segmentation; series deformable convolution

1. Introduction

Retinal blood vessel is the only human terminal vessel that could be observed in living body[1], and abnormal changes of its parameters are closely related to the disease onset. Retinal microvascular diameter abnormal changes are associated with diabetic retinopathy[2].The retinal vascular density in the outer macular area positively correlated with the degree of visual field defect in glaucoma, Vasospasm, narrowing of blood vessels and widening of arterial reflex zones. The cross compression of arteries and veins in the retina are related to hypertension and arteriovenous sclerosis[4]. Therefore, accurately segmenting retinal blood vessels and analyzing their morphological characteristics have important application values in the prevention, diagnosis, monitoring and management of various diseases, especially chronic diseases.

At present, retinal image segmentation methods can be divided into the retinal image segmentation unsupervised method (RISUM) and the retinal image segmentation supervised method (RISSM). Common RISUMs include RISUMs based on vascular tracking [5], mathematical morphology [6], active contour[7]and graph-theory[8]. RISUMs,using the inherent relationship between features to identify target blood vessels have the advantage of not having to train classifiers and adjust parameters, which leads to operation process simple. However, RISUMs have two disadvantages. Firstly, RISUMs usually assume that each pixel is independent for the vascular feature encoding which reduces the correlation between pixels. Secondly, RISUMs only use the pixel-level local information and ignore more important information such as the context information. The two disadvtangs above make the segmentation accuracy low for blood vessels.

The RISSMs are more reliable and stable than RISUMs[9]. RISSMs train classifiers that could distinguish blood vessels from the background by extracting various vessel features and learning label category information. Common RISSM include RISSM based on machine learning and RISSM based on deep learning. The RISSMs based on machine learning depend on the quality of artificial feature selection and can not adaptively extract the features of multi-scale, central reflex and geometric morphological changes of blood vessels,which lead to the problems of insufficient segmentation of small vessels and false segmentation of lesions.

In recent years, with the development of deep learning in computer vision, the great progress has been made in the fields of segmentation[10], etc. Supervised methods based on deep learning are also beginning to be applied in the field of retinal vessel segmentation.

Shelhamer et al. [11]proposed the fully convolutional network (FCN) to take advantage of the neighboring information for the structured prediction. The FCN could make good use of the adjacent pixel information with label correlation which is benifit for the segmentation accuracy of retinal blood vessels. In 2016, Fu et al. [12]used FCN to generate a vascular segmentation map to achieve better segmentation result. However, max-pooling operations in FCNs would reduce the localization accuracy and segmentation accuracy. To resolve the problems caused by max-pooling operation of FCNs, Ronneberger[13]et al. designed the U-Net based on the structure of the FCN. Wang et al. [14]used the U-Net for retinal blood vessel segmentation. The U-Net could propagate the context feature information of blood vessels from the U-Net down-sampling part to the U-Net up-sampling part.In this propagation process, the context feature information with different resolutions obtained from down-sampling part and the up-sampling part could be combined, which would improve the segmentation accuracy.

Laibacher et al. [15]proposed M2U-Net, in which the MobileNetV2 was added in the coding stage of U-Net and the bilinear sampling structure was added to replace the CNN of U-Net to reduce M2U-Net parameters. Zhuang et al.[16] proposed a LadderNet based on concatenated U-Net, which constructed multiple paths from input to output and made full use of the context information of pixels.

Compared with the previous methods, the above methods based on U-Net have excellent medical image segmentation performance, but still have the following problems. First, the low segmentation accuracy for complex morphological blood vessels. Due to the retinal blood vessels have complex curvature morphology, the above methods would lead to discontinuous segmentation, especially at highly complex vessel morphology during feature extraction. At the same time, the above methods use CNN to extract features, the amount of training data and the number of network layers would increase as the training progresses, which would lead to exploding gradient and vanishing gradient in the late training period. Second, the low segmentation accuracy for low contrast between blood vessels and background. In the retinal images, there is a low contrast between blood vessels and background noise, especially at the small vessels and at the lesions. The above methods would cause the loss of small vessels and mis-segmentation in the process of segmenting retinal blood vessels, which affects the segmentation accuracy.

To solve these problems, we propose a retinal vessel segmentation method with series deformable convolution and attention mechanism based on U-net structure (SDAU-Net). SDAU-Net improves the accuracy and sensitivity of small blood vessels segmentation, which also works well on lesion images and low-contrast images. The main contributions of our work are summarised in the following points.

To complex morphological blood vessels, we propose to use the series deformable convolution(SDC) instead of existing convolution of U-Net in SDAU-Net. The SDC could adaptively capture the features of various complex curvature morphological vessels and small vessels, which would not only improve the segmentation accuracy of high complexity vessels and small vessels, but also alleviate the problems of exploding gradient and vanishing gradient due to the excessive number of the network layers.

To low contrast between blood vessels and background, we propose to use the lightweight attention mechanism(LAM) and the dual attention mechanism(DAM) in SDAU-Net. LAM establishes the dependencies between every two features in the middle layer feature map to improve the utilization rate of the middle layer feature map. DAM uses its internal the two parallel attention models to captures the dependencies of spatial information and channel information of pixels respectively. Through the dependencies, LAM and DAM give greater weight to features beneficial to vessel segmentation, which could enhance the differentiation of vessels from the background and reduce the incidence of small vessel loss and mis-segmentation.

2. Method

As shown in Fig. 1, SDAU-Net a basic encoder and decoder network architecture, which have three specific innovations:

1. The CNN in U-Net is replaced by SDC. SDC is composed of a common convolution unit and two deformable convolution units, whose structure is similar to residual connections. SDC could improve the question of exploding gradient and vanishing gradient caused by the increase of network layers, strengthen feature propagation and increase feature utilization rate.

2. Two LAMs are used in the process of feature fusion between the encoder and the decoder at each layer to solve the problem that the traditional convolution cannot fully consider the global information.

3. Two DAMs are used in the each convolution layer of decoder module. DAM could effectively improve the segmentation accuracy of small vessels at the crossing vessels by combining the spatial information and channel information of the vascular features.

SDAU-Net encoder module consists of two sets of down-sampling layers and three sets of convolution layers. Each convolution layer consists of two series deformable convolution modules. The number of feature channels of the convolutional layers are 32, 64 and 128, respectively. The decoder module consists of two sets of up-sampling layers and two sets of convolution layers, in which each convolution layer consists of two SDCs, one LAM and one DAM. The number of feature channels in the convolution layer is 64 and 32, respectively. The decoder module adds a 1×1 common convolution unit at the end of the highest convolution layer to obtain the output.

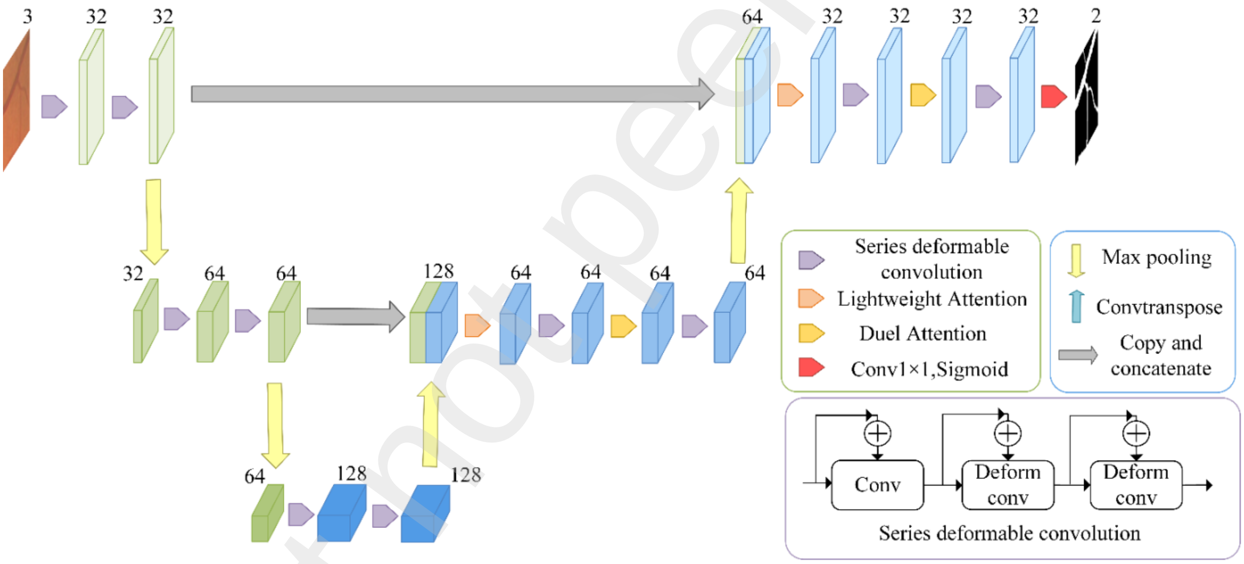

Fig. 1. The SDAU-Net architecture. To illustrate the attention mechanism, the feature map in the encoding process is represented by green and the feature map in the decoding process is represented by blue. Purple arrow with 3×3 and numbers (32,64,128) correspond to the convolution kernel size and the output channels. The size of the feature map before the first pooling is 48×48, the size of the feature map before the second pooling is 24×24 and the size of the feature map after the second pooling is 12×12. The number above the feature map represents the number of channels.

2.1. Series Deformable Convolution

We used SDC to replace the convolutional module in this paper. SDC is shown in Figure 2(a).

1)Series Structure: The series structure uses the cumulative result of the convolution result of the previous layer and the convolution result of this layer as input to the lower layer, which would enhance the transfer of features for efficient use of the output feature map. The series structure is defined as:

$$Input = x_1 \tag{1}$$

$$x_2 = F_1(x_1) + x_1 \tag{2}$$

$$x_3 = F_2(x_2) + x_2 \tag{3}$$

$$x_4 = F_2(x_3) + x_3 \tag{4}$$

$$Output = x_4 \tag{5}$$

Where x_i is the input feature mapping for layer i, $F_1(\cdot)$ is a standard convolution layer, and $F_2(\cdot)$ is a deformable convolution layer.

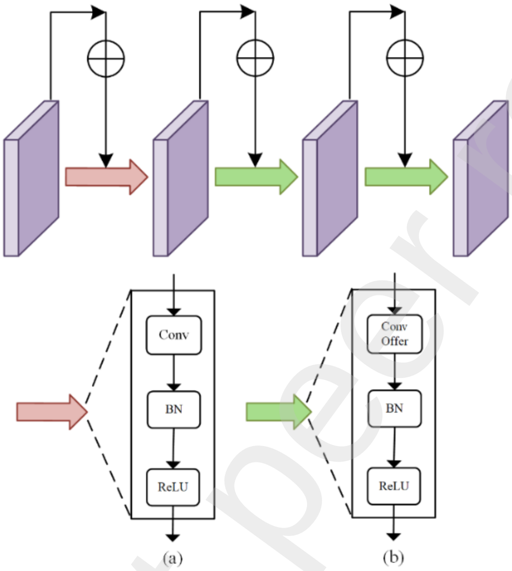

Fig. 2. SDC structure (a) Traditional convolution. (b) Deformable convolution.

2)Deformable Convolution: The feature extraction process for deformable convolution is shown in Figure 3. A learnable offset is added to the original sampling point, so that the shape and position of the convolution kernel could be dynamically adjusted according to the current vessel shape.

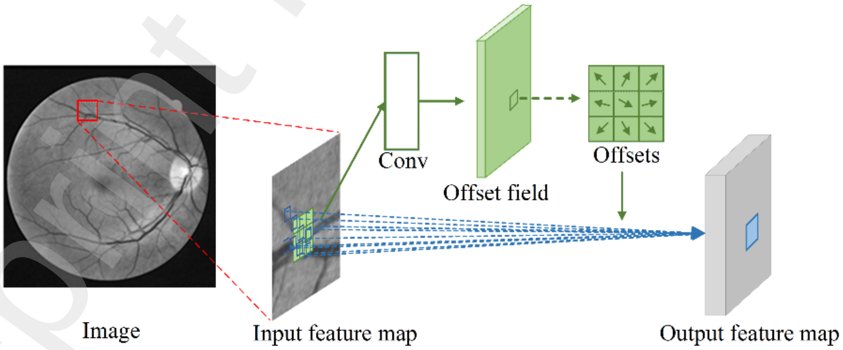

Fig. 3. Deformable convolution feature extraction process.

Take a convolution kernel size of 3×3 for example. As shown in FIG. 4(b), the standard convolution kernel regularly falls on a square region R on the fundus image to capture the vascular features in this region. R is defined as:

$$R=\{(-1,1),(-1,0),\dots,(0,1),(1,1)\} \quad (6)$$

So, each location t_0 from output feature map y is obtained as:

$$y(t_0)=\sum_{t_i\in R} w(t_i)\, f(t_0+t_i),\ i=1,2,\dots,N,\ N=|R| \quad (7)$$

Where f is the input feature map, w is the sample weight and t_i is the location of R.

The output corresponding to the deformable convolution is obtained as:

$$y(t_0)=\sum_{t_i\in R} w(t_i) f(t_0+t_i+\Delta t_i) \quad (8)$$

where, Δt_i is the variable offset. As shown in Fig. 4(c), Δt_i could be adjusted adaptively according to the structural characteristics of blood vessels in retinal images.

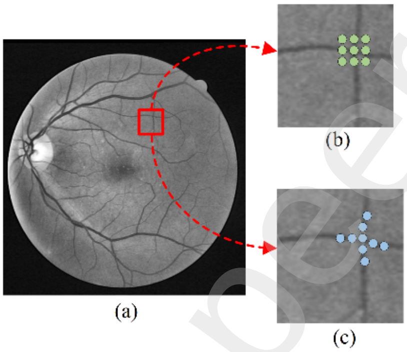

Fig. 4. Illustration of the sampling locations in 3×3 normal convolution and deformable convolution. (a) Pre-processed image. (b) Normal convolution. (c) Deformable convolution

2.2. Lightweight Attention Mechanism

A LAM is used in the feature fusion process between encoder and decoder, as shown in Fig. 5. LAM has the following three stages: feature extraction, attention map calculation and original feature enhancement.

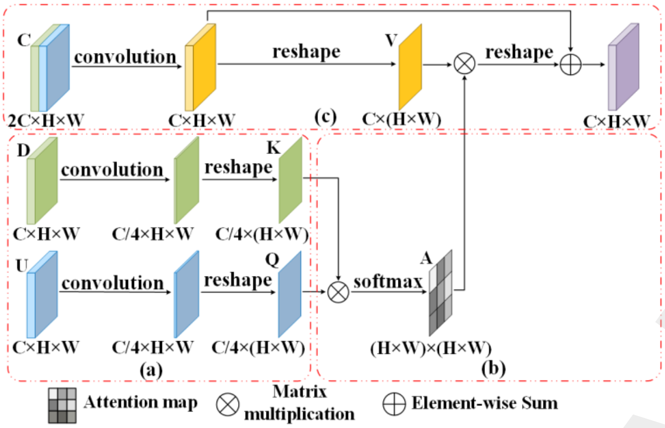

Fig. 5. Structure of the proposed lightweight attention. (a) Feature extraction. (b) Attention map calculation. (c) Original feature enhancement.

1) Feature Extraction: As shown Fig. 5(a), D and U are symmetrical features in the encoding and decoding process, $\{D,U\} \in R^{C\times H\times W}$, where C, H and W are the number of channels, height and width, respectively. The cascade feature $F \in R^{2C\times H\times W}$ is obtained by connecting D and U in the channel dimension. The original features D, U, and F are mapped into K, Q, and V spaces by convolution and reshape operation to complete the feature extraction.

2) Attention Map Calculation: As in Fig. 5(b), the similarity of matrix K and matrix Q is calculated by equation (9), in which the dot product method is used to calculate the similarity matrix f(·).

$$f(\bullet) = K^T Q \tag{9}$$

The f(·) is passed through by a softmax activation function to get the attention map $A \in R^{N\times N}$. The attention map describes the similarity between any two feature points in the K matrix and the Q matrix, which is defined as:

$$a_{ij} = \exp/\sum_{i=1}^{N} \exp(K_i \times Q_j) \tag{10}$$

Where a_{ij} is the influence between the i^{th} feature in the K matrix and the j^{th} feature in the Q matrix.

According to the above two steps, the connection between the global features could be established.

3)Original Feature Enhancement: As shown in Fig. 5(c), matrix V represents the feature mapping of the V space. The dot product of the matrix V and the attention map is used to obtain the enhanced features. Then, the enhanced features would be reshaped from two dimensions to three dimensions, whose dimension size is $R^{C\times H\times W}$. At last, the attention feature map is added to the original feature map. The above steps could be described by the following equation:

$$F_j = \gamma \sum_{i=1}^{N} (a_{ij} V_i) + X_j \tag{11}$$

Where F_j is the j^{th} feature generated by the attention module,γ is the coefficient of the attention features, which is used to measure the attention feature proportion in all features. γ which is initialized to 0 could be obtained by back propagation through learning.

2.3. Dual Attention Mechanism

As shown in Figure 6, a dual attention mechanism is used in each convolution layer of decoder, which is composed of position attention and channel attention.

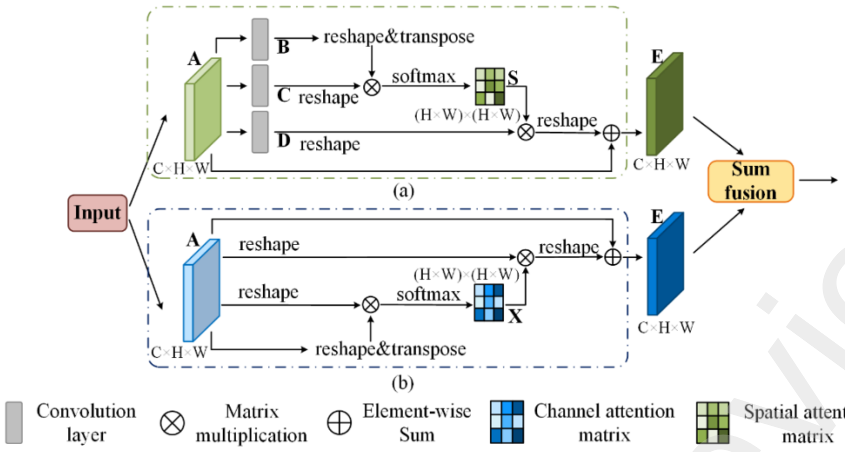

Fig. 6. Structure of the proposed lightweight attention. (a) Position attention. (b) Channel attention.

*1)Position Attention:*As shown as Fig. 6(a), the location attention is added to capture the spatial correlation between any two positions in the feature map. The specific implementation steps are as follows:

Firstly, the input feature map A obtains three identical feature maps B, C and D through convolution.

Secondly, the influence of the i^{th} position feature in B matrix on the j^{th} position feature in the C matrix is calculated by equation (10) to obtain a spatial attention map S, $S \in R^{N \times N}$.

Thirdly, the weighted sum of the features across all positions and original features is obtained equation (13).

$$E_j = \alpha \sum_{i=1}^{N} \left(S_{ij} D_i \right) + A_j \tag{12}$$

Where α is initialized to 0 and gradually assigned more weight through learning. E_j has a global contextual view and selectively aggregates contexts according to the spatial attention map. The similar semantic features achieve mutual gains to improve intra-class compact and semantic consistency.

*2)Channel Attention:*The structure of channel attention is shown in Fig.6(b). We use channel attention to explicitly model the interdependencies between channels by capturing channel dependencies between any two channel maps. Channel attention establishes the interdependence between channels by capturing the channel dependencies between any two channel mappings.

The original features $A \epsilon R^{C \times H \times W}$ are reshaped to $A \epsilon R^{C \times N}$, then x_{ji} would be obtained by equation (15).

$$x_{ji} = \exp(A_i \bullet A_j) / \sum_{i=1}^{C} \exp(A_i \bullet A_j) \tag{13}$$

Where x_{ji} is the influence between the i^{th} channel's and the j^{th} channel.

Then, the channel attention map X,$X \in R^{C \times C}$ could be obtained by a softmax activation function.

The final output $E \in R^{C \times H \times W}$ could be obtained by equation (15), which is defined as:

$$E_j = \beta \sum_{i=1}^{C} (x_{ij} A_i) + A_j \tag{14}$$

In equation (16), β is a scale parameter which could be gradually learned a weight from 0, the results of AX^T should be reshaped to $R^{C \times H \times W}$, the final output $E \in R^{C \times H \times W}$ could be obtained by performing an element-wise sum operation with A.

The Equation (16) shows that the final feature of each channel is a weighted sum of the features of all channels and original features, which models the long-range semantic dependencies between feature maps.

3. Databases and Evaluation Metrics

3.1. Databsets

There are four datasets in this paper, which are DRIVE, STARE, CHASE_DB1, and IOSTAR. The DRIVE dataset consists of 40 color fundus images with the image resolution of 565×584. It was divided into a training set and a test set. The training set and the test set each contain 20 images. The STARE dataset consists of 10 fundus images with lesions and 10 fundus images without lesions. The image resolution is 605 × 700. The CHASE_DB1 dataset consists of 28 color fundus images with the image resolution of 999 ×960. Compared with DRIVE and STARE datasets. The IOSTAR dataset were obtained by EasyScan based on the Scanning Laser Ophthalmoscopy (SLO) technique. The IOSTAR dataset has 30 images with a resolution of 1024×1024, and each image has manual segmentation map and key point coordinates.

3.2. Evaluation Metrics

The Se, Sp, Acc, and area under the ROC curve (AUC) are used to objectively evaluate the effect of retinal vessel segmentation.

$$Se = TP / (TP + FN) \tag{15}$$

$$SP = TN / (TN + FP) \tag{16}$$

$$ACC = (TP + TN) / (TN + TP + FN + FP) \tag{17}$$

Where TP, FP, TN and FN respectively represent true positive, false positive, true negative and false negative. The true positive indicates the number of pixels that correctly classify blood vessels, the false positive indicates the number of pixels that incorrectly classify background, the true negative indicates the number of pixels that correctly classify background, and the false negative indicates the number of pixels that incorrectly classify blood vessels.

4. Experiment

4.1. Experiment Platform

All experiments in this paper were conducted under the Tensorflow [44] and Keras [45] frameworks using Intel Xeone5-2678 V3 CPU and NVIDIA GeForce RTX 2080Ti GPU.

4.2. Image Preprocessikng and Data Enhancement

In order to improve the contrast between blood vessel and background, reduce image noise and improve uneven illumination, the original image is preprocessed as follows: 1) red-green channel fusion; 2) image normalization; 3) Contrast Limited Adaptive Histogram Equalization (CLAHE); 4) gamma correction.

We also carry out the patch-based segmentation and data augmentation for data enhancement to alleviate the insufffficiency of training data and avoid SDAU-Net overfitting. Firstly, we mirror flip 50% of the images and flip the remaining 50% left and right. After that, we subject all the flipped images to the affffine transformation of ±25% translation and ±10% rotation. Secondly, we locally twist the images by moving a single pixel in the distortion field with the intensity of 0.15. Each pixel moving range is 0.01 to 0.1 intensity. Thirdly, we blur each image by a Gaussian kernel and add with noise and brightness adjustment.

4.3. Experimental parameter setting

1)Dataset split: The official data split for the DRIVE dataset is adopted, which contains 20 training images for construction and 20 testing images for evaluation. Since no official data split is available for the other three datasets, the cross-validation method is adopted to split the datasets for a fair comparison experiment. The specific splits are the ten-fold cross-validation for STARE, four-fold cross-validation for CHASE_DB1 and five-fold cross-validation for IOSTAR. Hence, there are 18, 21, and 24 images for training and the remaining images for validation in the three datasets, respectively.

*2)Parameter setting:*In the experiment, 30000 patches from each image of DRIVE, STARE, CHASE_DB1 and IOSTAR are extracted to train SDAU-Net, the size of each patch is 48×48. We adopt the mini-batch Adam algorithm with a batch size of 16 as the optimizer with the default setting and set the maximum iteration number to 20. We adopt dynamic learning rate with the initial learning rate 0.001 , the scaling value 0.2 and the lower limit of the learning rate decrease 0.000001. The detection value is the loss of the verification set. The loss function is a cross entropy loss function.

5. Quantitative Analysis Results

5.1. SDAU-Net Performance

Table I shows the experiment results of SDAU-Net on the DRIVE, STARE, CHASE_DB1 and IOSTAR datasets.

Table 1.Performance Indicators comparision on the DRIVE, STare, CHASE_DB1 And IOSTAR Datasets

Dataset	Se	Sp	Acc	AUC
DRIVE	0.7955	0.9848	0.9682	0.9834
STARE	0.8973	0.9903	0.9833	0.9963
CHASE_DB1	0.8321	0.9825	0.9732	0.9858
IOSTAR	0.8842	0.9927	0.9848	0.9961

As shown in Table , Acc of SDAU-Net is higher than 0.9680 in all four datasets, which shows that SDAU-Net has universality. The Sp values of SDAU-Net in the four datasets are 0.9807, 0.9903, 0.9825 and 0.9927, respectively, which shows that SDAU-Net have a high recognition rate of retinal image background. The Se values of SDAU-Net in the four datasets are 0.8293, 0.8973, 0.8321 and 0.8842, respectively, which shows that SDAU-Net could extract small vascular images close to ground truth images. Among the four datasets, the Se value of SDAU-Net was the lowest on the DRIVE dataset and the highest on the STARE dataset. The reason for this result is that DRIVE is the dataset with the largest proportion of small vessels which are difficult to identify. On the contrary, the STARE dataset has relatively few manually annotated small vessels which are easy to identify.

5.2. Comparison With the State-of-the-Art-Methods

The comparison experiment results of SDAU-Net with existing methods on four datasets are shown in Table. 2 to Table 5, in which the best values are bolded.

Table 2. Comparsion experiment results on DRIVE dataset with other STATE-of-the-Art-Methods

	Method	Year	Se	Sp	Acc	AUC
Unsupervised methods	Wang et al.[22]	2013	-	0.9751	0.9461	0.9543
	Azzopardi et al.[23]	2015	0.7655	0.9751	0.9442	0.9614
	Zhang et al.[24]	2016	0.7743	0.9725	0.9476	0.9636
Supervised methods	Nicola et al.[25]	2016	0.7777	0.9702	0.9454	0.9597
	Oalando et al.[26]	2017	0.7897	0.9684	0.9454	0.9506
	Zhang et al.[27]	2017	0.7861	0.9712	0.9466	0.9703
Deep learning methods	Ronneberger et al.[13]	2015	0.7537	0.9820	0.9531	0.9738
	Yan et al.[28]	2018	0.7569	0.9813	0.9527	0.9738
	Gu et al.[29]	2019	0.8309	0.9747	0.9545	0.9777
	Li et al.[30]	2020	0.7921	0.9810	0.9568	0.9806
	Huang et al.[31]	2020	**0.8340**	0.9786	0.9560	0.9820
	Yang et al.[32]	2020	0.7735	**0.9845**	0.9573	0.9816
	Li et al.[33]	2020	0.7991	0.9813	0.9581	0.9823
	Wang et al.[34]	2020	0.7357	0.9760	0.9552	-
	Proposed method	2021	0.8293	0.9807	**0.9675**	**0.9832**

Table 3. Comparison experiment results on STARE dataset with other STATE-of-the-Art-Methods

	Method	Year	Se	Sp	Acc	AUC
Unsupervised methods	Wang et al.[22]	2013	-	-	0.9521	0.9682
	Azzopardi et al.[23]	2015	0.7716	0.9701	0.9497	0.9563
	Zhang et al.[24]	2015	0.7791	0.9758	0.9554	0.9758
Supervised methods	Nicola et al.[25]	2016	0.8046	0.9710	0.9543	0.9638
	Oalando et al.[26]	2017	0.7680	0.9738	0.9619	0.9570
	Zhang et al.[27]	2017	0.7882	0.9729	0.9547	0.9740
Deep learning methods	Ronneberger et al.[13]	2015	0.7924	0.9844	0.9541	0.9693
	Yan et al.[28]	2018	0.7581	0.9846	0.9612	0.9801
	Gu et al.[29]	2019	0.7841	0.9725	0.9583	0.9787
	Li et al.[30]	2020	0.8352	0.9711	0.9678	0.9875
	Huang et al.[34]	2020	0.8274	0.9528	0.9431	0.9758
	Yang et al.[31]	2020	0.8334	0.9897	0.9663	0.9734
	Li et al.[32]	2020	0.7715	0.9886	0.9701	0.9881
	Wang et al.[33]	2020	0.8186	0.9844	0.9673	0.9881
	Proposed method	2021	**0.8973**	**0.9903**	**0.9833**	**0.9963**

Table 4. Comparison of the proposed method on CHASE_DB1 dataset with other STATE-of-the-Art-Methods

	Method	Year	Se	Sp	Acc	AUC
Unsupervised methods	Azzopardi et al.[25]	2015	0.7585	0.9587	0.9387	0.9487
	Sohini et al.[35]	2015	0.7615	0.9575	0.9467	0.9623
	Zhang et al.[24]	2016	0.7626	0.9661	0.9452	0.9606
Supervised methods	Fraz et al.[36]	2012	0.7224	0.9711	0.9469	0.9712
	Oalando et al.[27]	2017	0.7565	0.9655	0.9467	0.9478

	Method	Year	Se	Sp	Acc	AUC
	Zhang et al.[28]	2017	0.7644	0.9716	0.9502	0.9706
Deep learning method	Ronneberger et al.[13]	2015	0.8288	0.9701	0.9578	0.9780
	Zhang et al.[37]	2018	0.7677	0.9825	0.9649	0.9839
	Yan et al.[29]	2018	0.7633	0.9809	0.9610	0.9780
	Fu et al.[17]	2019	0.7859	0.9822	0.9644	0.9834
	Jiang et.al[38]	2019	0.7686	**0.9861**	0.9662	0.9770
	Li et al.[31]	2020	0.7818	0.9819	0.9635	0.9810
	Proposed method	2021	**0.8321**	0.9825	**0.9732**	**0.9858**

Table 5. Comparison of the proposed method on IOSTAR dataset with other advanced methods

	Method	Year	Se	Sp	Acc	AUC
Unsupervised methods	Azzopardi et al.[21]	2015	0.7610	0.9670	0.9410	0.9550
	Abbasi-Sureshjani et al. [26]	2015	0.7863	0.9747	0.9501	0.9615
	Zhang[24]	2016	0.7545	0.9740	0.9514	0.9615
Deep Learning method	Ronneberger et al. [13]	2015	0.8044	0.9793	0.9675	0.9464
	Meyer et al[39]	2017	0.8038	0.9801	0.9695	0.9771
	Alom et al.[40]	2018	0.8042	0.9779	0.9652	0.9530
	Zhao et al.[41]	2018	0.7720	0.9670	0.9480	0.9600
	Srinidhi et al[42]	2018	0.8269	0.9669	0.9564	0.9663
	Gu et al.[30]	2019	0.8110	0.9749	0.9572	0.9658
	Fu et al.[17]	2019	0.8298	0.9832	0.9720	0.9504
	Mou et al.[43]	2020	0.8341	0.9831	0.9722	0.9758
	Li et al.[31]	2020	0.7322	0.9802	0.9544	0.9623
	Proposed method	2021	**0.8842**	**0.9927**	**0.9848**	**0.9961**

As shown in Table 2 to Table 5, Compared with the most advanced unsupervised learning methods and machine learning methods, the retinal blood vessels segmentation method based on deep learning has higher accuracy. On the DRIVE and IOSTAR datasets, which have a higher proportion of small vessels with complex morphology than the other two datasets, SDAU-Net achieves the best performance. This indicates that SDAU-Net has distinct advantage on the segmentation of small vessels with complex morphology. The STARE dataset and CHASE_DB1 dataset contain a large number of lesion abnormal images with uneven illumination, low contrast, and similar morphology of blood vessels and some pathological sites, which leads to difficulty in retinal blood vessel segmentation. On the STARE dataset, SDAU-Net has the highest values for the four evaluation metrics. On the CHASE_DB1 dataset, SDAU-Net has the highest values for the three evaluation metrics except for Sp, which is slightly lower than the method proposed by Jiang et al. This result indicates that SDAU-Net performs better in most cases and has strong feature extraction capability. In addition, Se represents the ability in distinguishing blood vessels from background. SDAU-Net has the highest Se value on both datasets, which indicates that SDAU-Net is optimal in solving problems such as uneven illumination, uneven contrast and mis-segmentation of vessels. The above experiment results show that SDAU-Net with SDC has the stronger ablity of adaptively extracting retinal blood vessel features for different thicknesses, high complexity blood vessels and small blood vessels. LAMs and DAMs of SDAU-Net could better distinguish the blood vessel from the background, effectively inhibit noise, and improve the accuracy of the blood vessel segmentation. The experimental results are consistent with the theoretical analysis.

5.3. Cross Dataset and Cross Modal Evaluation

To quantify the generalization and extendibility of the proposed model, the experiment of cross dataset and cross modality are conducted.

*1)Cross Dataset Experiment:*To verify generalization, cross-training experiment were conduct on DRIVE and STARE datasets. The experiment is divided into two stages, one stage is that SDAU-Net is trained on the DRIVE dataset and is used for segment on the STARE dataset, the other is that SDAU-Net is trained on the STARE dataset and is used for segment on the STARE dataset. Table 6 and Table 7 shows the cross dataset experiment results based of SDAU-Net and comparison methods. The segmented image of SDAU-Net is shown in Fig. 7.

Table 6. Cross-Training Assessment Based on DRIVE and STARE Datasets (Train on DRIVE and Test on STARE)

Model	Se	Sp	ACC	AUC
Yan et al.[44]	0.7219	0.9830	0.9580	0.9678
Jin et al.[45]	0.7000	0.9759	0.9474	0.9571
Proposed	**0.7298**	**0.9839**	**0.9646**	**0.9743**

Table 7. Cross-Training Assessment Based on DRIVE and STARE Datasets (Train on STARE and Test on DRIVE)

Model	Se	Sp	ACC	AUC
Yan et al.[44]	0.7014	0.9802	0.9444	0.9568
Jin et al.[45]	0.6505	0.9914	0.9481	0.9718
Proposed	**0.7077**	**0.9905**	**0.9658**	**0.9725**

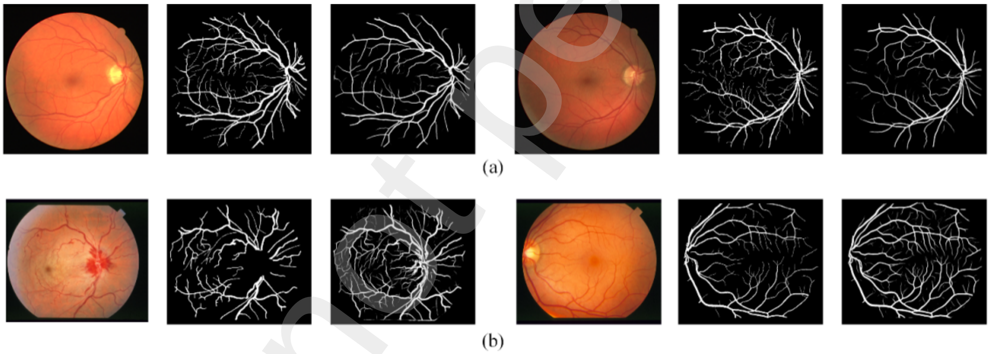

Fig. 7. The visualization results of cross dataset experiment on DRIVE and STARE datasets. From left to right are input image, ground truth image, segmented image. (a) The visualization results of training on STARE dataset and segmenting on DRIVE dataset. (b) The visualization results model training on DRIVE dataset and segmenting on STARE dataset.

As shown in Table 6 and Table 7, The experimental results of SDAU-Net in cross-dataset evaluation are better than the comparison methods, which shows that SDAU-Net has good generalization performance. Compared with Table 2 and Table 3, Sp, Acc and AUC in Table 6 and Table 7 have little difference, but the Se of Table 6 is significantly lower than the Se of Table 2 and Table 3. The reasons are analyzed as follows:As can be seen from Fig.7(a), the ground truth of STARE dataset contains mainly retinal thick vessels, when SDAU-Net is trained on STARE, the SDAU-Net parameters are fit for thick vessel segmentation. In contrast, the DRIVE dataset contains more fine vessels compared to the STARE dataset, which leads to a relatively low Se.

2) Cross Modality Experiment: To verify extendibility on different imaging modalities datasets, cross-training evaluations were conducted on DRIVE and IOSTAR datasets. The DRIVE dataset and the IOSTAR dataset are datasets constructed in two different imaging modalities, belonging to the fundus color camera imaging modalities and the scanning laser ophthalmoscope (SLO) imaging modalities, respectively. At present, few studies have evaluated images of cross-training performance from different imaging modalities of retinal images. Table 8 shows cross-training evaluation results for datasets from SLO-based imaging modalities. The experiment was pre-trained on the DRIVE and tested on the IOSTAR.

Table 8. Cross-Training Assessment Based on DRIVE and STARE Datasets (Train on DRIVE and Test on STARE)

Model	Se	Sp	Acc	AUC
HAnet	0.7903	0.9757	0.9668	0.9763
SDAU-Net	**0.8823**	**0.9869**	**0.9838**	**0.9924**

As shown in Table 8, SDAU-Net has better extendibility the compared method.

5.4. Blood Vessel Analysis

1)High Complexity Blood Vessel Image Segmentation Analysis: In order to further illustrate the segmentation effect of SDAU-Net for small vessels with complex morphology, comparative experimental analysis are conducted on DRIVE and IOSTAR datasets which contains a large number of small vessels with complex morphology. Fig.10 shows the segmentation results for different models in the DRIVE dataset and the IOSTAR dataset. In Fig.10, (a) and (b) are the raw image and manual annotation respectively, (c), (d) and (e) are the segmentation results of Jin et al.[45], Li et al.[30] and Wu et al. [46], respectively, (f) is the segmentation result of SDAU-Net.

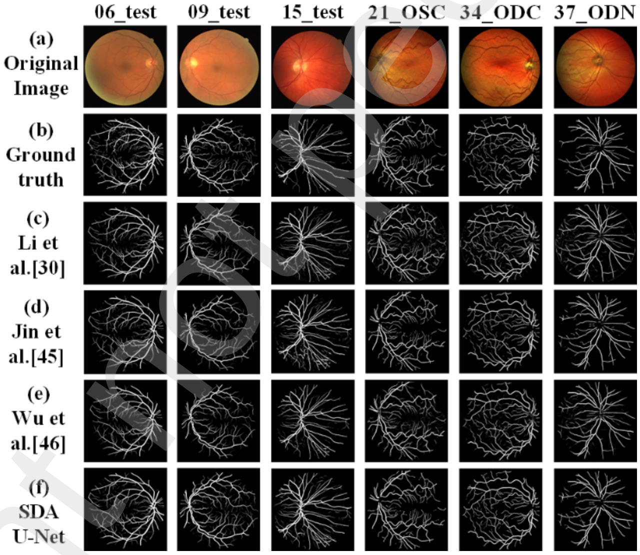

Fig. 8. Visual comparison between SDAU-Net and state-of-the-art networks for retinal vessel segmentation on DRIVE and IOSTAR datasets. The first three retinal images of the first line belong to the DRIVE dataset and the last three retinal images the first line belong to the IOSTAR dataset.

As shown in Fig.8:

The segmentation result of 09_test showed that SDAU-Net achieves optimal segmentation results at vessel intersections and small vessel endpoints compared with the methods of Jin et al., Li et al. and Wu et al.. The segmentation results of 15_test showed that SDAU-Net has good segmentation result for the detailed parts of some high-complexity vascular regions, such as the intersection of multiple vessels, compared with the method of Li et al.

The segmentation results of 34_ODC and 37_ODN showed that compared with the methods proposed by Jin et al. and Wu et al., SDAU-Net segmented small vessels better, with few vessel breaks and stronger robustness.

The above analysis shows that SDAU-Net has a good segmentation result for small blood vessels with high morphological complexity. This indicates that SDC could effectively capture the information of the complex structure of blood vessels, which improves the segmentation accuracy. Meanwhile, the attention mechanism could put more weight on the blood vessels and increase the contrast between the blood vessels and the background, which solve the problem of blood vessel segmentation fracture well.

*2) Detail Analysis of High Complexity Blood Vessel Image Segmentation:*The segmentation results of complex blood vessel regions in DRIVE and IOSTAR datasets are selected for analysis, as shown in Fig. 9.

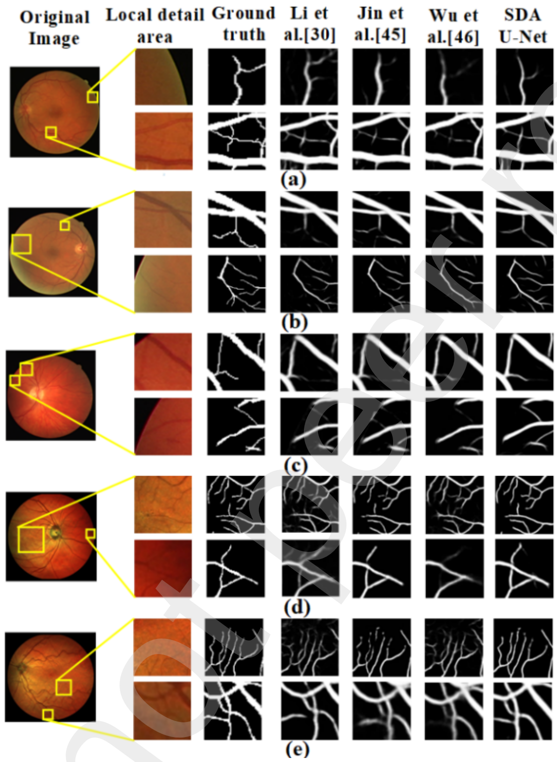

Fig. 9. Magnified visual comparison of SDAU-Net with the state-of-the-art segmentation method on DRIVE and IOSTAR dataset. The yellow boxed patches represents the comparison area.

As can be seen from (a), (b) and (c) in Fig. 9:

The models proposed by Jin et al. , Li et al. and Wu et al. could only extract rough blood vessel contour information near the intersection of multiple blood vessels and could not accurately segment the small blood vessels attached to rough blood vessel. SDAU-Net could captureretinal blood vessels of various diameter and successfully segment the blood vessels which seem to be entangled but actually separated. The experimental results in Fig. 11 show that SDAU-Net is better than the other three methods in the case of complex interlaced vascular trees and could achieve better results.

As could be seen from Fig. 9(d) and (e):

The method proposed by Jin et al. and Wu et al. could only segment the general contour of thin blood vessels, and the segmentation results of the method proposed by Li shows a large number of vessel rupture. The two attention mechanisms of SDAU-Net make the model pay more attention to the thin vessels information of highly complex, reducing the interference of background on blood vessel segmentation, and improving the segmentation accuracy.

3) Analysis of Low Contrast Blood Vessel Image Segmentation: To further illustrate the effectiveness of SDAU-Net for vessel segmentation of lesion images and low-contrast images, the comparative analysis is conducted on STARE and CHASE_DB1 datasets which contains more lesion images and low-contrast images in this section, as shown in Fig. 10.

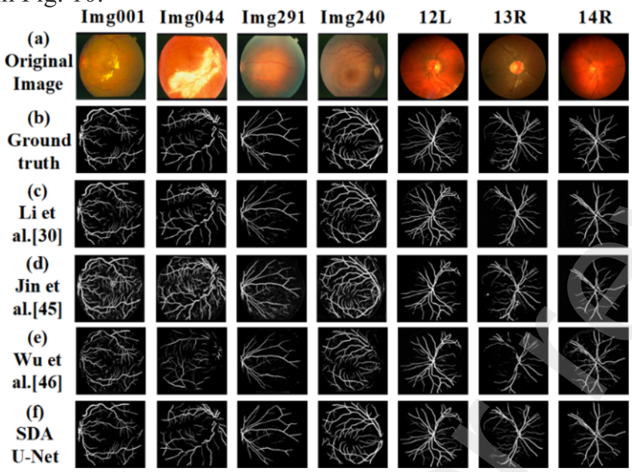

Fig. 10. Visual comparison of our proposed SDAU-Net with the state-of-the-art segmentation methods for retinal vessel segmentation on STARE and CHASE_DB1 datasets. The first four retinal images belong to the STARE dataset and the last three retinal images belong to the CHASE_DB1 dataset.

As shown in Fig.10:

By observing the original images, labels and segmentation results of Img044 and Img291, it could be seen that the large number of bright spots and exudates in the original image make the segmentation of vessels more difficult and lead to discontinuous segmentation results. The methods proposed by Jin et al. and Li et al. mistakenly segment exudate as blood vessels. Although the methods proposed by Wu et al. avoided the occurrence of mis-segmentation, the segmentation result still shows the insufficient segmentation of small blood vessels. Compared with the methods proposed by Jin et al. , Li et al. and Wu et al. , SDAU-Net could effectively eliminate the influence of bright spot background and accurately segments the discontinuous blood vessels.

By observing the original image, labels and segmentation results of 13_R, it could be seen that the original image is dark and uneven in illumination, which lead to inaccurate segmentation of the crossed vessels at the optic disc and the small vessels around the optic disc not easily to be segmented. However, SDAU-Net could accurately segment the small blood vessels on the left side of the optic disc. Compared with the method proposed by Wu et al, the segmentation results of SDAU-Net on the cross vessels at the optic disc do not show the phenomenon of vascular rupture.

4)Detail Analysis of Low Contrast Blood Vessel Image Segmentation: To specifically illustrate the segmentation effect of SDAU-Net on lesion images, retinal images with low contrast, the STARE and CHASE_DB1 datasets were selected for amplification and comparison, as shown in Fig. 11.The segmentation results of complex blood vessel regions in DRIVE and IOSTAR datasets are selected for analysis, as shown in Fig. 11.

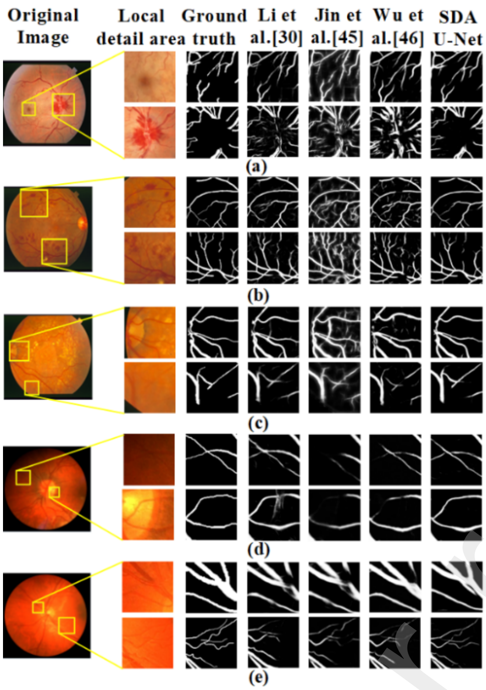

Fig. 11. Magnified visual comparison of SDAU-Net with the state-of-the-art segmentation methods for retinal vessel segmentation on STARE and CHASE_DB1 dataset. The yellow-boxed patches represents the comparison area we selected.

As can be seen from Fig. 11(a) and (b):

The diseased plaques result in significant differences between adjacent areas, creating a situation where one part is bright and the adjacent part is dark, which increases the difficulty of vessel segmentation. The methods proposed by Jin and Li et al. show fine vessel rupture at the intersection of blood vessels. The method proposed by Wu et al. is easy to misidentify the noise of bright spots as blood vessels, which leads to inaccurate blood vessel segmentation. The methods proposed by Jin et al. , Li et al. , and Wu et al. , all mis-segment hard exudate as blood vessels. However, SDAU-Net uses the attention mechanism to increase the feature differentiation between blood vessels and lesion sites, which reduce the influence of lesion regions on vessel segmentation. In addition, the attention mechanism enhances the features favorable to vessel segmentation to avoid the occurrence of small vessel breakage and vessel mis-segmentation.

As can be seen from Figs. 11(c), (d) and (e):

In the vicinity of the optic disc region, affected by uneven illumination, the results of the methods proposed by Li et al. and Wu et al. show vessel rupture at the vascular branches, vascular merger of different degrees. The results of the method proposed by Jin et al. and Li et al. show optic disc false segmentation of different degrees. While SDAU-Net could not only avoid vessel rupture, but also segment vessels of different thicknesses well.

5.5. Ablation Experiment Based on Drive Dataset

The ablation experiments are conducted on different modules of SDAU-Net on DRIVE dataset. U-NET is the baseline. The experimental results are shown in Table 9.

Table 9. Ablation experment for SDAU-Net on DRIVE dataset

Class	The evaluation Metrics			
	Se	Sp	Acc	AUC
SubNet_1	0.7537	0.9820	0.9531	0.9738
SubNet_2	0.7850	0.9830	0.9667	0.9801
SubNet_3	0.7833	**0.9851**	0.9678	0.9808
SubNet_4	**0.7955**	0.9848	**0.9682**	**0.9834**

As shown in Table 9, SubNet_1 represents the baseline U-NET. SubNet_2 represents the result of SDC joining SubNet_1. SubNet_3 represents the result of LAM joining SubNet_2. SubNet_4 represents the result of DAM joining SubNet_3.

As shown in Table 9, SubNet_1 already has good results for retinal vessel segmentation using only U-Net, and the various metrics are better than those in the literature, but the Se and Acc values are low and need further improvement. SubNet_2 has a better effect than SubNet_1. The four performance metrics of blood vessel segmentation are increased by 0.0313, 0.001, 0.0136 and 0.0063, respectively. The Sp, Acc, and AUC of SubNet_3 are improved by 0.0009, 0.0011, and 0.0007 respectively, in which the Acc value increased significantly. In SubNet_4, In particular, the segmentation effect for some small blood vessels is further improved.

6. Conclusion

SDAU-Net is proposed to solve the problem of insufficient segmentation of small vessels with complex curvature and difficult segmentation of vessels due to uneven brightness of retinal image background. Comparative experimental results show that SDAU-Net is superior to the compared method in the segmentation of thin blood vessels in retinal images with complex curvature morphology and lesions. The experimental results of cross datasets and cross modalities show that the proposed method has good generalization ability and scalability. The results of ablation experiments show that the network design of this method is reasonable.

References

[1]W. J. Lucas et al., “The plant vascular system: evolution, development and functions,” Journal of integrative plant biology., vol. 55, no. 4, pp. 294--388, 2013.
[2]M. K. Ahsan et al., “Ocular fundus abnormalities in pre-dialytic chronic kidney disease patients,” Journal of Biosciences and Medicines, vol. 7, no. 11, pp. 20, 2019.
[3]T.Jurecka, Z. Batkova, and J. Ventruba, ”Macular edema after an uncomplicated cataract surgery,” World journal of diabetes., vol. 7, no. 19, pp. 483--514, 2016.
[4]J. A. Paniagua, “Insulin resistance and dysfunctional adipose tissue determine the different components of metabolic syndrome,” World journal of diabetes., vol. 7, no. 19, pp. 483, 2016.
[5]E. Bekkers, R. Duits, T. Berendschot, and B. T. H. Romeny, “A multi-orientation analysis approach to retinal vessel tracking,” Journal of Mathematical Imaging and Vision., vol. 49, no. 3, pp. 583--610, 2014.
[6]A. M. Mendonca, and A. Campilho, “Segmentation of retinal blood vessels by combining the detection of centerlines and morphological reconstruction,” IEEE Trans. Med.Imag., vol. 25, no. 9, pp. 1200--1213, 2006.
[7]Y. Q. Zhao, X. H. Wang, X. F. Wang, and F. Y. Shih, “Retinal vessels segmentation based on level set and region growing,” Pattern Recognition., vol. 47, no. 7, pp. 2437--2446, 2014.
[8]X. Xu, M. Niemeijer, Q. Song, M. Sonka, M. K. Garvin, J. M. Reinhardt, and M. D. Abramoff, “Vessel boundary delineation on fundus images using graph-based approach,” IEEE Trans. Med.Imag., vol. 30, no. 6, pp. 1184--1191, 2011.
[9]Y. Guo and Y. Peng, “BSCN: bidirectional symmetric cascade network for retinal vessel segmentation,” BMC medical imaging., vol. 20, no. 1, pp. 1--22, 2020.

[10]S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," Neural Inf Ormation Processing Systems.$, vol. 28, pp. 91–99, 2015.
[11]E. Shelhamer, J. Long,and T. Darrell, "Fully convolutional networks for semantic segmentation," in Proc. CVPR, 2015, PP. 3431--3440.
[12]H. Fu, Y. Xu, and D. W. K. Wong, "Retinal vessel segmentation via deep learning network and fully-connected conditional random fields," in Proc. ISBI, 2016, pp. 698--701.
[13]O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in arXiv preprint arXiv:1505.04597$, 2019.
[14]W. Xiancheng et al., "Retina blood vessel segmentation using a U-net based Convolutional neural network," in Proc. ICDS, 2018, pp. 8--9.
[15]T. Laibacher, T. Weyde, and S. Jalali, "M2u-net: Effective and efficient retinal vessel segmentation for real-world applications," in Proc. CVPR, 2019.
[16]J. Zhuang et al., "LadderNet: Multi-path networks based on U-Net for medical image segmentation," in arXiv preprint arXiv:1810.07810$, 2018.
[17]J. Fu, J. Liu, H. Tian, Y. Li, Y. Bao, Z. Fang, and H. Lu, "Dual attention network for scene segmentation," in Proc. CVPR, 2019, pp. 3146–3154.
[18]J. Staal, M. D. Abràmoff, M. Niemeijer, M. A. Viergever, amd B. Van Ginneken, "Ridge-based vessel segmentation in color images of the retina" in IEEE Trans. Med. Imag., vol. 23, pp.501-509, 2004.
[19]A. Hoover, V. Kouznetsova, and M. Goldbaum, "Locating blood vessels in retinal images by piecewise threshold probing of a matched filter response," in IEEE Trans. Med. Imag., vol. 19, pp. 203-210, 2000.
[20]C. G. Owen et al., "Measuring retinal vessel tortuosity in 10-year-old children: validation of the computer assisted image analysis of the retina (CAIAR) program," in Investigative ophthalmology & visual science., vol. 50, pp.2004-2010, 2009.
[21]S. Abbasi-Sureshjani, I. Smit-Ockeloen, J. Zhang, and B. T. H.Romeny, "Biologically-inspired supervised vasculature segmentation in SLO retinal fundus images," in Proc. ICIAR, 2015, pp. 325–334.
[22]Y. Wang, G. Ji, P. Lin, and E. Trucco, "Retinal vessel segmentation using multiwavelet kernels and multiscale hierarchical decomposition," Pattern Recognition., vol. 46, no. 8, pp. 2117--2133, 2015.
[23]N. Strisciuglio, G. Azzopardi, M. V ento, and N. Petkov. "Supervised vessel delineation in retinal fundus images with the automatic selection of b-cosfire filters," Machine Vision and Applications., vol. 27, no. 8, pp. 1137–1149, 2016.
[24]J. Zhang, B. Dashtbozorg, E. Bekkers, J. P . Pluim, R. Duits, and B. M. ter Haar Romeny, "Robust retinal vessel segmentation via locally adaptive derivative frames in orientation scores," IEEE Trans. Med. Imag., vol. 35, no. 12, pp. 2631–2644, 2016.
[25]N. Strisciuglio, G. Azzopardi, M. V ento, and N. Petkov, "Supervised vessel delineation in retinal fundus images with the automatic selection of B-COSFIRE filters," Machine Vision and Applications., vol. 27, no. 8, pp. 1137--1149, 2016.
[26]J. I. Orlando, E. Prokofyeva, and M. B. Blaschko, "A discriminatively trained fully connected conditional random field model for blood vessel segmentation in fundus images," IEEE Trans. Biomed. Eng., vol. 64, no. 1, pp. 16--27, 2017.
[27] J. Zhang, Y. Chen, E. Bekkers, M. Wang, B. Dashtbozorg, and B. M. ter Haar Romeny, "Retinal vessel delineation using a brain-inspired wavelet transform and random forest," Pattern Recognition., vol. 69, pp. 107--123, 2017.
[28]Z. Yan, X. Yang, and K.T. Cheng, "Joint segment-level and pixel-wise losses for deep learning based retinal vessel segmentation," IEEE Trans. Biomed. Eng., vol. 65, no. 9, pp. 1912--1923, 2018.
[29]Z. Gu et al., "Ce-net: Context encoder network for 2d medical image segmentation," Neurocomputing., vol. 38, no. 10, pp. 2281--2292, 2019.
[30]X. Li, Y. Jiang, M. Li, and S. Yin, "Lightweight attention convolutional neural network for retinal vessel image segmentation," IEEE Trans. Ind. Informat., vol. 17, no. 3, pp. 1958--1967, 2020.
[31]T. Yang, T. Wu, L. Li, and C. Zhu, "SUD-GAN: deep convolution generative adversarial network combined with short connection and dense block for retinal vessel segmentation," Journal of digital imaging., vol. 33, no. 4, pp. 946--957, 2020.
[32]L. Li, M. Verma, Y. Nakashima, H. Nagahara, and R. Kawasaki, "Iternet: Retinal image segmentation utilizing structural redundancy in vessel networks," in Proc. WACV, 2015, pp. 3656--3665.
[33]D. Wang, A. Haytham, J. Pottenburgh, O. Saeedi, and Y. Tao, "Hard attention net for automatic retinal vessel segmentation," IEEE J. Biomed. Health Inform., vol. 24, no. 12, pp. 3384--3396, 2020.
[34]L. Huang, and F. Liu, "Retinal vessel segmentation using simple SPCNN model and line connector," The Visual Computer., pp. 1--14, 2020.
[35]S. Roychowdhury, D. D. Koozekanani, and K. K. Parhi, "Iterative vessel segmentation of fundus images," IEEE Trans. Biomed. Eng., vol. 62, no. 7, pp. 1738–1749, 2015.
[36]M M Fraz, P Remagnino, and A Hoppe, et al., "An ensemble classification-based approach applied to retinal blood vessel segmentation". IEEE Trans. Biomed. Eng., vol. 59, no. 9, pp. 2538--2548, 2012.
[37]Y. Zhang, and A. C. S. Chuang, "Deep supervision with additional labels for retinal vessel segmentation task," in Proc. MICCAI, 2018, pp. 83--91.
[38]Z Jiang $et al., "Retinal blood vessel segmentation using fully convolutional network with transfer learning," Computerized Medical Imaging and Graphics., vol. 68, pp. 1--15, 2018.

[39]M. I. Meyer, P. Costa, A. Galdran, A. M. Mendonc¸a, and A. Campilho, “A deep neural network for vessel segmentation of scanning laser ophthalmoscopy images,” in Proc. ICIAR, 2017, pp. 507--515.
[40]M. Z. Alom, M. Hasan, C. Yakopcic, T. M. Taha, and V. K. Asari, “Recurrent Residual Convolutional Neural Network based on U-Net (R2U-Net) for Medical Image Segmentation,” in arXiv preprint arXiv:1802.06955, 2018.
[41]Y. Zhao et al., “Automatic 2-D/3-D vessel enhancement in multiple modality images using a weighted symmetry filter,” IEEE Trans. Med. Imag., vol. 37, no. 2, pp. 438--450, 2017.
[42]C. L. Srinidhi, P. Aparna, and J. Rajan “A visual attention guided unsupervised feature learning for robust vessel delineation in retinal images,” Biomedical Signal Processing and Control., vol. 44, pp. 110--126, 2018.
[43]L. Mou et al., “CS2-Net: Deep learning segmentation of curvilinear structures in medical imaging,” Medical image analysis., vol. 67, pp. 101874, 2021.
[44]Z. Yan, X. Yang, and K. T. Cheng “A three-stage deep learning model for accurate retinal vessel segmentation,” IEEE J. Biomed. Health Inform., vol. 23, no.4, pp. 1427--1436, 2018.
[45]Q. Jin, Z. Meng, T. D. Pham, Q. Chen, L. Wei, and R. Su “DUNet: A deformable network for retinal vessel segmentation,” Knowledge-Based Systems., vol. 178, pp. 149--162, 2019.
[46]Y. Wu, Y. Xia, Y. Song, Y. Zhang, and W. Cai, “Multiscale network followed network model for retinal vessel segmentation,” in Proc. MICCAI, 2018, pp. 119--126.